# What Financial Statements Can and Cannot Reveal About Monetary Transmission to Corporate Balance Sheets: A Five-Country Emerging-Market Test

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**Abstract**

Firms holding floating-rate debt bear the brunt of monetary tightening—a finding built almost entirely on supervisory credit registries. Most emerging markets have none; they have published financial statements. Can these substitute? We develop a measurement framework in which exposure is estimated rather than observed, and test its predictions on harmonised panels of listed non-financial firms in five emerging markets—Türkiye, Vietnam, Poland, Israel and South Korea—over 2010–2025 (2021–2025 for the latter three), spanning tightening doses from Türkiye's 33.5 percentage points to Vietnam's 0.8. The price of debt proves recoverable: a simple implied borrowing cost tracks each country's policy path, and its change correlates with tightening dose at $r = 0.996$ across the sample, with pass-through of roughly 35% in Türkiye and coverage halving. Exposure does not. Split-half reliabilities of $-0.075$ and $-0.082$ in the two countries with decade-long history imply an attenuation factor of zero, Türkiye's inflation-accounting restatement enters as an exposure-by-post interaction fixed effects cannot absorb,

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and pooling all five countries the exposure-by-dose interaction is null on interest coverage and profitability ($p = 0.93$, $p = 0.42$) while significant only on the price measure already shown to move with dose mechanically. The contractual object these designs require is absent from the accounting record.
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## 1. Introduction

A firm that borrows at a floating rate feels a policy tightening within weeks. A firm that borrowed long and fixed does not feel it until it refinances. This asymmetry is simple, intuitive, and has become the organising principle of a productive literature on how monetary policy reaches the corporate sector. Ippolito et al. (2018) identify a *floating rate channel* in matched loan-level records, showing that firms with unhedged floating-rate bank debt cut investment and employment more sharply after tightening. Şengül and Çinko (2026), studying the June 2023 policy reversal in Türkiye, apply a two-part Double Machine Learning design to a large administrative dataset and find that while the average effect is modest, sensitivity concentrates in firms with weak internal risk ratings, low liquidity and high exposure shares, with exporters and manufacturers relatively insulated.

What these studies share is not merely a question but an input: a credit registry recording, loan by loan, whether the contract reprices. Such registries exist in relatively few jurisdictions and are available to researchers in fewer still. What almost every market does have is a body of audited, published financial statements. The question this paper takes seriously—and, we argue, the field has not asked carefully enough—is whether that accounting record can stand in for the registry.

The stakes are practical. If statements suffice, the literature's geographic reach expands by an order of magnitude. If they do not, then a body of work is being built on designs that cannot deliver what they promise, and it is worth establishing precisely where the failure occurs and what remains possible.

### *1.1. Approach*

We proceed in two stages. Section 2.2 develops a measurement framework in which the exposure variable is *estimated* rather than observed, and derives what that implies for the difference-in-differences estimand. The framework delivers four propositions, each of which generates a sharp, falsifiable prediction about statistics we can compute. Sections 3.1–5.4 take those predictions to data.

The empirical design is comparative and, where the data permit, decade-long. Vietnam and Türkiye anchor the study: they share the structural features that make the floating-rate channel plausible—bank-dominated corporate finance, thin corporate bond markets, heavy reliance on short-maturity credit, substantial listed non-financial sectors—and diverge sharply in monetary experience. Türkiye executed one of the decade's most severe tightenings, from 9.0% to 47.5%, under inflation exceeding 60%; Vietnam tightened modestly in late 2022 and reversed within a year. Both countries carry the history needed to test the identifying assumption directly, back to 2010–11.

A two-point contrast, however sharp, is still only two points, and a referee is right to ask whether either country's null is idiosyncratic. We therefore extend the design to three further emerging markets—Poland, Israel and South Korea—spanning tightening doses of 5.0, 4.65 and 2.5 percentage points and chosen, among a broader set of markets assembled during scoping, for the largest and most reliable firm counts.[1] Across the five countries the change in implied borrowing cost correlates with tightening dose at $r = 0.996$ (Section 5.3), and the exposure-by-dose interaction is null on distress and profitability outcomes regardless of how severely a country tightened. The two-country design supplies the depth—a decade of history, a testable pre-trend, an inflation-accounting stress test; the five-country extension supplies breadth, showing the conclusion is not an artefact of which two countries were chosen.

This dose gradient is the paper's instrument throughout. It imposes a discipline a single-country study cannot: a measurement strategy that works should register a stronger response where tightening was more severe, in

[1] Two further markets, Brazil and Chile, were assembled on the same basis and are not carried into the reported five-country test. Chile (7.25pp dose, 80 firms) sits squarely on the same dose-response line as the five reported countries; its exclusion is for parsimony alone. Brazil (5.75pp dose, 215 firms) is the one clear outlier in the full assembled set, with a change in implied borrowing cost of only 0.48 percentage points against a pass-through ratio roughly a quarter of every other country's—$0.48/5.75$ against a range of 0.16 to 0.62 elsewhere. We report this so the reader can judge for themselves: the five-country correlation below is not conditioned on having found and discarded an inconvenient point after the fact, but its absence should be read as a reduction in scope, not as evidence that Brazil disproves the pattern.

proportion. A method that fails to register the gradient, or that finds a firm-level exposure effect that itself fails to scale with dose, has failed a test it would never have faced with one country alone.

We are conscious that five countries is not many, and Section 5.3 returns to this directly: with so few points a very high correlation is not, on its own, strong statistical evidence, and we read it qualitatively rather than as a precisely estimated slope.

Our diagnostic standard throughout is *validation against an external benchmark*. We do not ask whether a measure is statistically well behaved; we ask whether it recovers something independently known. For the price of debt, the benchmark is the published policy path. For exposure, it is the firm's own subsequent change in borrowing cost. For the panel, it is an independent commercial rendering of the same filings.

### *1.2. What we find*

The constructive half of the answer concerns the *price* of corporate debt. A firm's implied borrowing cost—interest expense divided by average debt—aggregates to a series reproducing each central bank's policy path without being shown it. The Turkish series peaks in 2024, the year policy peaks; the Vietnamese series traces the 2011 inflation crisis, the long easing through 2015, and the 2023 tightening. This establishes that ordinary filings carry a legible signal of monetary transmission.

Two results follow. We quantify pass-through: a 38.5 percentage point Turkish policy move produces a 13.4 point move in realised borrowing costs, roughly 35%, while median interest coverage halves from 3.36 to 1.66. Section 2.2 shows how to read that number structurally as an effective repricing share. Second, the same diagnostic exposes a data hazard: a widely used Turkish vendor field labelled *Financial Expenses* bundles foreign-exchange revaluation with interest, running roughly two and a half times the true figure, so measures built on it peak in the lira crises of 2018 and 2021 rather than at the 2024 policy peak.

The cautionary half concerns *exposure*, and here theory and evidence coincide exactly. Proposition 1 shows that using an estimated exposure attenuates the estimand by the reliability ratio $\lambda$. Proposition 2 shows $\lambda$ is identified by the split-half correlation of independently estimated exposures. We measure that correlation at $-0.075$ in Türkiye and $-0.082$ in Vietnam, implying $\hat{\lambda} \approx 0$: the estimand is attenuated to zero regardless of the true effect, and a null is therefore uninformative about the underlying parameter. Proposition 4 derives the panel length required for feasible estimation and shows annual accounting data falls short by roughly an order of magnitude. Proposition 6 derives the in-sample fit expected under the null, $k/(T-1) \approx 0.40$, against observed medians of 0.49 in Vietnam and 0.66 in Türkiye—not an exact match, but the same order of magnitude as chance, which is the point: a two-parameter fit on six annual observations looks moderately good whether or not the relationship is real.

Layered on top is a problem specific to high-inflation settings. Proposition 7 shows that when non-monetary assets are restated by a vintage-dependent factor while monetary debt is not, the distortion enters as an exposure-by-post interaction that two-way fixed effects cannot absorb. Proposition 8 shows that ratios of two flows escape it. We demonstrate the former empirically: a standard specification returns highly significant wrong-signed coefficients on a monotonically growing post-treatment path.

### *1.3. Why a null of this kind is informative*

Null results invite the suspicion of low power, and we address it directly. Over the analysis three separate significant results appeared and each was shown to be an artefact: a Turkish effect of +0.886 ($p = 0.045$) that vanished when the sample doubled; the FX results, which were accounting restatement; and the final flow-based coefficients, which were pre-existing trend. Each was caught by a diagnostic—sample expansion, an accounting-mechanics argument, an extended event window—that a conventional specification would not have run. A design generating three false positives before generating a null is informative about the design.

### *1.4. Contribution*

To the transmission literature we delimit where statement-based designs can substitute for registry data, and supply a validated price measure that survives. To empirical work on Türkiye we document two hazards—one in a vendor field, one in the accounting standard—threatening a broad class of post-2022 studies. To research on Vietnam we introduce a firm-level statement panel not previously used for such questions. Methodologically, we offer split-half reliability and first-stage prediction as cheap, decisive tests that any paper using an estimated firm characteristic as a regressor should report, and we give them a formal justification.

## 2. Related literature and analytical framework

### 2.1. Related literature

#### 2.1.1. Firm heterogeneity in monetary transmission

That monetary policy affects firms unequally is long established. Gertler and Gilchrist (1994) documented that small manufacturing firms bear a disproportionate share of the adjustment following tightening, and Bernanke and Gertler (1995) formalised the credit channel through which balance-sheet strength mediates the response. Kashyap and Stein (2000) established the bank lending channel using cross-sectional variation in bank liquidity.

The modern literature has sharpened the relevant margin. Ottonello and Winberry (2020) show that firms with low default risk respond *most* to monetary shocks, because they face flatter marginal cost curves for external finance. Cloyne et al. (2023) find that younger firms without dividend payments drive the investment response. Ozdagli (2018) shows that financial frictions shape the stock price reaction to policy, and Greenwald (2019) identifies an interest coverage channel operating through debt covenants—directly relevant to our outcome set, since coverage is one of the few distress measures that survives Proposition 8.

Two strands bear closely on our design. The first concerns *contractual* repricing: Ippolito et al. (2018) identify the floating rate channel using loan-level data, and Auer et al. (2019) trace international transmission through banks in small open economies. The second concerns *maturity*: Almeida et al. (2012) exploit quasi-random variation in whether long-term debt happened to mature just before the 2007 crisis, a design that isolates rollover exposure cleanly; Jungherr et al. (2024) show maturity structure materially shapes transmission; He and Xiong (2012) provide the theoretical link between rollover and credit risk; and Demirgüç-Kunt and Maksimovic (1999) document how institutions shape maturity choice across countries.

Our contribution is not to this literature's substance but to its transportability. Every study cited uses either supervisory data or a market with rich contractual disclosure. We ask what remains when neither is available, and Section 2.2 shows the answer is not simply "a noisier version of the same design".

### *2.1.2. Firm-level evidence on Türkiye and comparable markets*

A body of firm-level work situates our exercise in the Turkish setting. Bilgin et al. (2023) study Borsa Istanbul firms directly and, importantly for us, exploit the *maturity breakdown* of bank credit in explaining corporate performance—establishing both that maturity structure is economically consequential in this market and that it is recoverable from disclosure. Sugozu et al. (2025) apply machine-learning methods to credit risk in Turkish participation and conventional banks. On the macro-financial side, Çitçi and Kaya (2023) analyse exchange-rate uncertainty and the connectedness of inflation in Türkiye, and Akbal and Civcir (2026) the asymmetric response of the lira to capital flows and sovereign spreads. Together these establish the two forces our measurement must separate—interest rates and the currency—as first-order in this economy.

A parallel literature examines firm financing structure in comparable markets: Çam and Özer (2022) on how country governance shapes capital structure and investment financing, Karakoç (2022) on the corporate growth–trade credit relationship across a panel of countries, and Huynh (2024) on banking uncertainty and trade credit. What these share with our setting is a reliance on published statements rather than supervisory records, which is precisely why the measurement limits we establish bear on them.

Our reading of this literature is that it has been appropriately cautious in what it asks of accounting data—studying maturity composition, credit type and financing structure, all of which are disclosed—and that the floating-rate question is different in kind because its treatment variable is not.

*2.1.3. Currency mismatch in emerging markets*

Because Türkiye's episode combines tightening with severe depreciation, the FX-mismatch literature bears directly on our measurement problem. Eichengreen and Hausmann (1999) framed the "original sin" of borrowing in foreign currency. The empirical record is more nuanced than the framing suggests: Bleakley and Cowan (2008) find that firms holding dollar debt did not systematically underperform during depreciations, because they tended to hold dollar revenues too, while Aguiar (2005) documents contractionary balance-sheet effects in Mexico. Kalemli-Ozcan et al. (2016) reconcile these by showing that the damaging combination is unhedged FX debt together with a banking crisis restricting credit supply.

Bruno and Shin (2015b,a) establish the risk-taking channel linking exchange rates, bank leverage and global liquidity; Du and Schreger (2022) connect sovereign risk, currency risk and corporate balance sheets; Acharya et al. (2024) show how impaired credit allocation distorts inflation dynamics. For Türkiye, Akbal and Civcir (2026) examine the asymmetric response of the lira to capital flows and sovereign spreads.

This literature matters to us twice over. Substantively it explains why exporters may be insulated—a mechanism Şengül and Çinko (2026) invoke directly, noting that exporters generate foreign-currency revenues and benefit from exchange-rate pass-through. Methodologically it explains why the bundled financial-expense field of Section 4.1 is so damaging: in a currency crisis revaluation losses dwarf interest, and a measure conflating them attributes currency effects to monetary policy.

#### *2.1.4. Accounting measurement under inflation*

Our identification problem is at bottom an accounting one, and there is a literature on it that monetary economics rarely cites. Gordon (2001) examines the value relevance of historical cost, price level and replacement cost accounting in Mexico; Barniv (1999) studies inflation-adjusted versus historical-cost earnings under hyperinflation; Davis-Friday and Gordon (2005) analyse how the valuation roles of book value, earnings and cash flows shift during a macroeconomic shock; and Konchitchki (2011) demonstrates that nominal reporting has real consequences even at moderate inflation. De Franco et al. (2011) show that statement comparability is itself economically consequential.

The consistent message is that when general price levels move sharply, the mapping from economic reality to reported figures changes, and changes *differently across firms.* Proposition 7 gives this literature what we believe is a novel application: an accounting transition coinciding with a treatment date and interacting with the treatment variable is not a nuisance to be absorbed but a threat to identification.

### *2.1.5. Measurement error and generated regressors*

Our exposure results connect to a developed econometric literature. Pagan (1984) established the consequences of using a regression-generated variable as a regressor in a second stage, which is precisely our situation. Griliches and Hausman (1986) show that panel transformations frequently *amplify* rather than attenuate measurement error—directly relevant, since first-differencing in (6) removes the fixed effect but magnifies the noise-to-signal ratio. Classical attenuation biases coefficients toward zero, exactly the pattern our null exhibits, which is why we regard a first-stage validity test as obligatory rather than optional.

Proposition 5 extends this in a direction we have not seen applied to firm-level exposure: separating estimation error from genuine impermanence of the underlying characteristic. The distinction is consequential because the two failure modes call for opposite responses.

## *2.2. An analytical framework for estimated exposure*

This section formalises what is at stake when the exposure variable of a difference-in-differences design is estimated from accounting data rather than observed contractually. The framework is deliberately spare, and each result generates a statistic we compute in Sections 4.1–5.1.

### *2.2.1. Setup*

Index firms by $i = 1, \ldots, N$ and years by $t = 1, \ldots, T$. Let $\pi_t$ denote the policy rate and $f_t$ the log exchange rate. Let $P_t = \mathbf{1}\{t \geq t^*\}$ be the post-treatment indicator, with $t^*$ the tightening date.

Each firm has a latent *repricing exposure* $\theta_i$, defined as the share of its debt whose contractual rate resets within the period. The object of interest is the effect of tightening on a distress outcome $Y_{it}$, mediated by exposure:

$$Y_{it} \;=\; \beta\left(\theta_i P_t\right) \;+\; \alpha_i \;+\; \lambda_t \;+\; \varepsilon_{it}, \qquad \mathbb{E}[\varepsilon_{it} \mid \theta_i, P_t] = 0, \tag{1}$$

with firm effects $\alpha_i$ and year effects $\lambda_t$. If $\theta_i$ were observed, $\beta$ would be identified by the two-way within estimator. The registry literature is, in effect, the case in which $\theta_i$ is observed.

Our situation is that $\theta_i$ is not disclosed. We observe instead a proxy

$$\widehat{\theta}_i \;=\; \theta_i + u_i, \qquad u_i \perp (\theta_i, \varepsilon_{it}), \qquad \mathrm{Var}(u_i) = \sigma_u^2, \;\; \mathrm{Var}(\theta_i) = \sigma_\theta^2. \tag{2}$$

Define the *reliability ratio*

$$\lambda \;\equiv\; \frac{\sigma_\theta^2}{\sigma_\theta^2 + \sigma_u^2} \;\in [0, 1]. \tag{3}$$

*2.2.2. Attenuation of the treatment estimand*

**Proposition 1** (Attenuation)**.** *Let $\widehat{\beta}$ be the two-way within estimator of* (1) *obtained by replacing $\theta_i$ with $\widehat{\theta}_i$. Then*

$$\operatorname*{plim}_{N\to\infty} \;\widehat{\beta} \;=\; \lambda\,\beta. \tag{4}$$

*Proof.* Write the regressor as $X_{it} = \widehat{\theta}_i P_t$ and let $\tilde{X}$ denote its residual after projecting out firm and year effects. Because $P_t$ varies only over time and $\widehat{\theta}_i$ only across firms, the within transformation yields $\tilde{X}_{it} = (\widehat{\theta}_i - \bar{\theta})(P_t - \bar{P})$. Using (2) and $u_i \perp \theta_i$,

$$\mathrm{Cov}(\tilde{Y}, \tilde{X}) = \beta\,\sigma_\theta^2\;\mathrm{Var}(P), \qquad \mathrm{Var}(\tilde{X}) = (\sigma_\theta^2 + \sigma_u^2)\;\mathrm{Var}(P).$$

The ratio is $\lambda\beta$. $\square$

Proposition 1 is the classical attenuation result (Pagan, 1984) transposed to an interaction term, and it has an uncomfortable implication: a statistically insignificant $\widehat{\beta}$ is consistent with *any* value of $\beta$ once $\lambda$ is small. A null is therefore uninterpretable without an estimate of $\lambda$.

#### *2.2.3. Identifying the reliability ratio*

The natural objection is that $\lambda$ is unobservable because $\theta_i$ is. It is not, provided two independent proxies can be constructed.

**Proposition 2** (Split-half identification)**.** *Let* $\widehat{\theta}_i^{(1)} = \theta_i + u_i^{(1)}$ *and* $\widehat{\theta}_i^{(2)} = \theta_i + u_i^{(2)}$ *be estimated on disjoint subsamples, with* $u^{(1)} \perp u^{(2)}$ *and* $\mathrm{Var}(u^{(1)}) = \mathrm{Var}(u^{(2)}) = \sigma_u^2$*. Then*

$$\mathrm{Corr}\big(\widehat{\theta}^{(1)}, \widehat{\theta}^{(2)}\big) \;=\; \frac{\sigma_\theta^2}{\sigma_\theta^2 + \sigma_u^2} \;=\; \lambda. \tag{5}$$

*Proof.* $\mathrm{Cov}(\widehat{\theta}^{(1)}, \widehat{\theta}^{(2)}) = \mathrm{Var}(\theta) = \sigma_\theta^2$ since the errors are mutually independent and independent of $\theta$. Each variance is $\sigma_\theta^2 + \sigma_u^2$. □

**Corollary 3.** *If the split-half correlation is not bounded away from zero, then* $\operatorname{plim} \widehat{\beta} = 0$ *irrespective of* $\beta$*, and the estimated treatment effect carries no information about the structural parameter.*

This is, to our knowledge, an underused device. The split-half correlation is trivial to compute whenever exposure is estimated from a time series, and by Proposition 2 it *is* the attenuation factor rather than merely a diagnostic correlated with it. Section 4.2 reports values of $-0.075$ and $-0.082$.

#### *2.2.4. Feasibility: how long a panel is required?*

Corollary 3 raises the question of whether a longer panel would rescue the design. It can be answered analytically.

Suppose exposure is estimated firm by firm from the first-differenced regression

$$\Delta r_{it} \;=\; a_i \;+\; \theta_i\,\Delta\pi_t \;+\; \gamma_i\,\Delta f_t \;+\; \eta_{it}, \qquad \mathrm{Var}(\eta_{it}) = \sigma_\eta^2, \tag{6}$$

where $r_{it}$ is the firm's realised borrowing cost. With $T$ usable observations per firm, sample variance $s_\pi^2$ of $\Delta\pi_t$, and correlation $\rho$ between $\Delta\pi_t$ and $\Delta f_t$, the sampling variance of the OLS estimate is

$$\mathrm{Var}(\widehat{\theta}_i) \;=\; \frac{\sigma_\eta^2}{T\,s_\pi^2\,(1-\rho^2)} \;\equiv\; \sigma_u^2. \tag{7}$$

**Proposition 4** (Feasibility condition)**.** *Attaining reliability* $\lambda \geq \lambda^*$ *requires*

$$T \;\geq\; \frac{\lambda^*}{1-\lambda^*}\cdot\frac{\sigma_\eta^2}{s_\pi^2\,(1-\rho^2)\,\sigma_\theta^2}. \tag{8}$$

*Proof.* Substitute (7) into (3) and solve the inequality $\sigma_\theta^2/(\sigma_\theta^2+\sigma_u^2) \geq \lambda^*$ for $T$. $\square$

Every quantity on the right is estimable, so (8) can be taken to data. When we do so in Section 4.2 the condition is comfortably satisfied: sampling noise implies $\lambda \approx 0.99$ in Vietnam and 0.78 in Türkiye, and a conventional power calculation would have judged the design feasible at the available panel length.

That verdict is contradicted by the split-half correlations, which are zero or negative. The contradiction is informative, and resolving it requires enriching (2).

*2.2.5. Permanent, transitory and sampling components*

Suppose the quantity a firm-level regression recovers is not a fixed trait but a window-specific realisation,

$$\theta_i^{(w)} \;=\; \theta_i \;+\; \nu_i^{(w)}, \qquad \mathrm{Var}(\nu_i^{(w)}) = \sigma_\nu^2, \tag{9}$$

where $\theta_i$ is a persistent characteristic and $\nu_i^{(w)}$ a transitory component that is real within window $w$—the firm genuinely did reprice that way over those years—but does not carry across windows. The estimated slope is then $\widehat{\theta}_i^{(w)} = \theta_i + \nu_i^{(w)} + u_i^{(w)}$.

**Proposition 5** (Three-way decomposition)**.** *Under* (9) *with* $\nu$*,* $u$ *mutually independent and independent of* $\theta$*,*

$$\mathrm{Var}(\widehat{\theta}_i^{(w)}) = \sigma_\theta^2 + \sigma_\nu^2 + \sigma_u^2, \qquad \mathrm{Corr}\big(\widehat{\theta}^{(1)}, \widehat{\theta}^{(2)}\big) = \frac{\sigma_\theta^2}{\sigma_\theta^2 + \sigma_\nu^2 + \sigma_u^2}, \tag{10}$$

*so the split-half correlation identifies the* permanent *share alone, and*

$$\sigma_\nu^2 \;=\; \mathrm{Var}(\widehat{\theta})\big(1 - \mathrm{Corr}\big) \;-\; \sigma_u^2. \tag{11}$$

*Moreover, Proposition 1 continues to hold with* $\lambda$ *equal to this permanent share, since only* $\theta_i$ *is stable enough to interact with a treatment occurring after the estimation window.*

*Proof.* The variance decomposition is immediate from independence. The covariance across windows is $\mathrm{Var}(\theta) = \sigma_\theta^2$ because both $\nu$ and $u$ are window-specific. Rearranging gives $\sigma_\nu^2$. $\square$

Proposition 5 separates two failure modes that are easily conflated. *Imprecision* ($\sigma_u^2$ large) is remediable by more data. *Impermanence* ($\sigma_\nu^2$ large) is not: no amount of additional data makes a transitory quantity a usable predetermined regressor. The empirical decomposition in Section 4.2 attributes 99.4% of Vietnamese and 78.0% of Turkish dispersion in estimated slopes to the transitory component and none to the permanent one—so the binding problem is impermanence, and quarterly estimation would not solve it.

*2.2.6. In-sample fit is not evidence*

A researcher inspecting (6) would observe a respectable $R^2$ and might conclude the specification has content. It does not.

**Proposition 6** (Null benchmark for fit)**.** *Under the null that neither regressor enters* (6)*, with $k$ regressors excluding the intercept and $T$ observations, the coefficient of determination satisfies*

$$R^2 \sim \mathrm{Beta}\big(\tfrac{k}{2},\ \tfrac{T-k-1}{2}\big)\,, \qquad \mathbb{E}\big[R^2\big] \;=\; \frac{k}{T-1}. \tag{12}$$

*Proof.* Standard: under the null the explained and residual sums of squares are independent $\chi^2$ variates with $k$ and $T-k-1$ degrees of freedom, so their normalised ratio is Beta distributed with the stated parameters, whose mean is $k/(T-1)$. □

With $k = 2$ and a median of $T = 6$ usable observations, $\mathbb{E}[R^2] = 0.40$. Any reported fit near this value is uninformative. Section 4.2 reports an observed Vietnamese median of 0.396.

*2.2.7. Inflation restatement as an exposure-by-post confound*

We now formalise the accounting problem described in Section 3.1. Let $A_{it}$ denote true non-monetary assets and $D_{it}$ monetary debt. Under a general price-level restatement beginning at $t_0$, non-monetary items are indexed by a factor depending on their *vintage*, while monetary items are not:

$$A^*_{it} \;=\; \Phi_{it}\, A_{it}, \qquad D^*_{it} \;=\; D_{it}, \qquad \log \Phi_{it} \;=\; \big(1+\delta_i\big)\, c_t\, \mathbf{1}\{t \geq t_0\}, \tag{13}$$

where $c_t$ is the common cumulative price-index component and $\delta_i$ is a firm-specific loading determined by the vintage profile of the firm's asset base, normalised so that $\mathbb{E}[\delta_i] = 0$.

**Proposition 7** (Non-absorbability)**.** *Let reported log leverage be* $\ell^*_{it} = \log D^*_{it} - \log A^*_{it}$ *and consider the two-way within regression of* $\ell^*_{it}$ *on* $\theta_i P_t$ *with* $P_t = \mathbf{1}\{t \geq t_0\}$*. Then*

$$\operatorname{plim}\ \widehat{\beta} \;=\; \beta \;-\; \bar{c}\,\frac{\operatorname{Cov}(\theta_i, \delta_i)}{\operatorname{Var}(\theta_i)}, \qquad \bar{c} \equiv \mathbb{E}\big[c_t \mid t \geq t_0\big], \tag{14}$$

*so the estimator is inconsistent whenever* $\operatorname{Cov}(\theta_i, \delta_i) \neq 0$*, and the bias is not removed by firm or year effects.*

*Proof.* Substituting (13), $\ell^*_{it} = \ell_{it} - c_t\mathbf{1}\{t \geq t_0\} - \delta_i c_t \mathbf{1}\{t \geq t_0\}$. The second term varies only with $t$ and is absorbed by $\lambda_t$. The third term varies across firms and only after $t_0$; it is therefore collinear with the regressor $\theta_i P_t$ to the extent that $\delta_i$ covaries with $\theta_i$, and projecting it onto $\theta_i P_t$ yields the stated bias. Firm effects $\alpha_i$ cannot absorb it because it is zero before $t_0$ and non-zero after. □

Proposition 7 is the paper's central identification result and the reason we regard the Turkish post-2022 window as hazardous. The condition $\operatorname{Cov}(\theta_i, \delta_i) \neq 0$ is not a knife-edge case: $\delta_i$ depends on asset vintage, which correlates with capital intensity, sector and firm age, each of which correlates with financing structure and hence with any balance-sheet exposure measure. The usual defence—country-by-year fixed effects—absorbs $c_t$ exactly and $\delta_i c_t$ not at all.

*2.2.8. Flow ratios are invariant*

The corresponding positive result identifies what survives.

**Proposition 8** (Flow invariance)**.** *Suppose income-statement items accruing within period* $t$ *are restated by a common factor* $\psi_t$ *that does not depend on* *i. Then for any two flows* $X_{it}, Z_{it}$ *the reported ratio is exactly invariant,* $X^*_{it}/Z^*_{it} = X_{it}/Z_{it}$*; and for the implied borrowing cost* $r_{it} = |\mathrm{Int}_{it}|/\bar{D}_{it}$*, which divides a restated flow by an unrestated monetary stock,*

$$\log r^*_{it} \;=\; \log\psi_t \;+\; \log r_{it}, \tag{15}$$

*so that* $\log r^*_{it}$ *is contaminated only by an additive time effect, fully absorbed by* $\lambda_t$*.*

*Proof.* Immediate from $X^* = \psi_t X$, $Z^* = \psi_t Z$ and $D^* = D$. $\square$

**Remark 1.** Proposition 8 is why our outcome set is restricted to the EBIT margin, interest coverage, the log implied rate and log debt growth, and why the Altman $Z''$ score, leverage, equity ratios and return on assets are discarded: each of the latter contains a non-monetary stock and is exposed to Proposition 7. It also refines the specification: the implied rate is invariant *in logs*, not in levels, and we use it accordingly.

*2.2.9. Reading pass-through structurally*

Finally, the framework gives the aggregate pass-through estimate an interpretation. Let the firm's realised average cost be a maturity-weighted average of past marginal rates,

$$r_{it} \;=\; \sum_{s\geq 0} \omega_{is}\, \iota_{t-s}, \qquad \sum_{s\geq 0} \omega_{is} = 1, \tag{16}$$

where $\iota_t$ is the marginal contract rate and $\omega_{is}$ the share of debt priced $s$ periods ago. Then $\partial r_{it}/\partial\iota_t = \omega_{i0} \equiv m_i$, the share repricing within the period, and aggregating, $\Delta\bar{r}/\Delta\bar{\iota} = \bar{m}$.

**Corollary 9.** *The aggregate pass-through coefficient identifies the mean effective repricing share $\bar{m}$ of the corporate debt stock.*

Section 4.1 estimates $\bar{m} \approx 0.35$ for Türkiye against a median short-term debt share of 0.63, implying that even nominally short debt repriced only partially—evidence, though not proof, of quantity rationing or subsidised credit substituting for full price adjustment.

## 3. Institutional setting and data

### *3.1. Institutional setting*

#### *3.1.1. Two monetary regimes*

The Turkish tightening is the sharper of the two by an order of magnitude. After a period of unorthodox easing the one-week repo rate stood at 9.0% at end-2022. Following the June 2023 reversal it reached 42.5% by end-2023, peaked at 47.5% during 2024, and stood at 38.0% at end-2025. Consumer price inflation exceeded 60% at its height and the lira depreciated from roughly 18.7 per US dollar at end-2022 to 43.0 at end-2025.

Vietnam's experience is far milder. The State Bank raised policy rates in late 2022 amid stress in the domestic corporate bond market and reversed through 2023. The average bank lending rate rose from 8.0% in 2022 to 9.3% in 2023—against 17.0% at the peak of the 2011 inflation episode, a useful within-country benchmark for what a severe Vietnamese tightening looks like.

The comparison is therefore not treatment versus control but large dose versus small dose, and the appropriate prediction is proportional rather than binary.

*3.1.2. The Turkish accounting transition*

Türkiye met the IFRS criteria for a hyperinflationary economy and was classified as such from April 2022, with local inflation-adjustment requirements following for statements through end-2023. Under the relevant standard, entities reporting in the currency of a hyperinflationary economy restate their statements in the measuring unit current at the balance sheet date, and comparatives are restated on the same basis.

The mechanics are those assumed in (13). Non-monetary items—property, plant and equipment, inventories, equity—are restated by a general price index applied from each item's acquisition date, so the effective factor depends on vintage. Monetary items—debt, cash, receivables—are already in period-end purchasing power and are not restated. Income-statement items are restated to period-end purchasing power by a factor common across firms within a period, which is the condition of Proposition 8.

*3.2. Data*

*3.2.1. Sources and construction*

Two countries anchor the study and supply its decade-long identification tests: Vietnam and Türkiye, described below. Three further countries—Poland, Israel and South Korea—extend the 2021–2025 window across a range of monetary tightening between Türkiye and Vietnam. These three are built from Yahoo Finance directly: ticker universes are enumerated per exchange via the TradingView screener, restricted to non-financial sectors, and statements retrieved as in the validation exercise of Section 4. Yahoo retains only four to five annual periods, so these three support the aggregate dose-response test of Section 5.2 but not the decade-long event-study test

Section 5.2 also requires for Türkiye and Vietnam; and by the feasibility condition of Section 2.1, a four-year window is in any case well short of what firm-level repricing-sensitivity estimation needs, so no attempt is made to extend the beta estimation of Section 4 to them.

The Vietnamese panel derives from an item-level dataset of statements filed by companies listed on the Ho Chi Minh City and Hanoi exchanges, distributed through a public research repository in long format with standardised item codes. Pivoting to firm-year observations yields 9,893 firm-years for 693 firms over 2011–2025, with non-missing coverage of 94.5% for total assets, 94.7% for total debt, 91.2% for interest expense, 93.2% for EBIT and 96.8% for equity.

The Turkish panel derives from a brokerage service redistributing filings made to the Public Disclosure Platform. One request returns the balance sheet, income statement, cash-flow statement and supplementary items with bilingual labels for four annual periods. The resulting panel covers 585 firms over 2010–2025 and carries two items with no Vietnamese counterpart: export sales and net foreign-currency position.

Both panels are restricted to non-financial firms, since banks, insurers and brokers file on incompatible templates. We require strictly positive total debt and assets, the transmission channel being undefined otherwise. Table 1 reports firm counts by year: Turkish coverage rises as the exchange admits new listings over the sample, while Vietnamese coverage is near-complete from 2015. Because identification uses within-firm variation with firm fixed effects, this entry does not bias the estimates, though earlier years rest on a smaller Turkish cross-section.

Table 1: Firms per year

| Year | Türkiye | Vietnam | Year | Türkiye | Vietnam |
|---|---|---|---|---|---|
| 2015 | 349 | 648 | 2021 | 551 | 691 |
| 2016 | 358 | 663 | 2022 | 568 | 692 |
| 2017 | 370 | 670 | 2023 | 585 | 693 |
| 2018 | 428 | 682 | 2024 | 584 | 692 |
| 2019 | 467 | 686 | 2025 | 582 | 689 |
| 2020 | 508 | 690 | | | |

### *3.2.2. Descriptive statistics*

Table 2 summarises the two anchor panels over 2015–2025.

Table 2: Descriptive statistics, 2015–2025

| | Vietnam | | | | Türkiye | | | |
|---|---|---|---|---|---|---|---|---|
| Variable | $n$ | Mean | Median | SD | $n$ | Mean | Median | SD |
| Debt / assets | 5,708 | 0.250 | 0.225 | 0.170 | 4,780 | 0.228 | 0.195 | 0.190 |
| Short-term debt share | 5,708 | 0.709 | 0.852 | 0.323 | 4,783 | 0.627 | 0.647 | 0.284 |
| Implied borrowing cost | 5,693 | 0.063 | 0.063 | 0.032 | 3,870 | 0.312 | 0.252 | 0.219 |
| Interest coverage | 5,340 | 9.225 | 3.290 | 15.698 | 4,673 | 2.680 | 1.068 | 6.502 |
| Return on assets | 5,698 | 0.065 | 0.053 | 0.065 | 4,783 | 0.080 | 0.070 | 0.105 |

Leverage is comparable across markets at 20–25% of assets, and both corporate sectors fund themselves predominantly at short maturity; the Vietnamese median short-term share of 0.85 speaks to the rollover-risk framing of He and Xiong (2012). The contrast in distress is stark: median Turkish interest coverage is 1.07 against Vietnam's 3.29, placing the median Turkish firm close to the covenant-relevant threshold Greenwald (2019) identifies.

### 3.2.3. *Cross-validation against an independent source*

Because the subject is measurement, we test rather than assume our panels. Table 3 reports the reconciliation.

Table 3: Reconciliation with an independent source

| | Vietnam | | Türkiye | |
|---|---|---|---|---|
| Variable | Corr. | Median diff. | Corr. | Median diff. |
| Total assets | 1.0000 | 0.0000 | 0.9635 | 0.0000 |
| Revenue | 0.9999 | 0.0000 | 0.9431 | 0.0000 |
| Total debt | 0.9906 | 0.0000 | 0.8934 | −0.0780 |
| Equity | 0.9956 | −0.0447 | 0.8412 | −0.0003 |
| EBIT | 0.7657 | +0.2339 | 0.8499 | +0.0798 |
| Interest expense (abs.) | 0.9913 | 0.0000 | **0.4950** | **−0.5955** |

Total assets and revenue reconcile at a median relative difference of zero in both countries, though the correlation is tighter in Vietnam (1.0000, 0.9999) than Türkiye (0.9635, 0.9431). Equity differs by about four percent in Vietnam and negligibly in Türkiye, in each case because the aggregator reports parent-only equity where the national source consolidates minority interest; EBIT differs by 23% in Vietnam and 8% in Türkiye because the aggregator derives it from pretax income plus interest rather than reported operating profit. All of the above are definitional. The interest-expense discrepancy in Türkiye is not, and Section 4.1 is devoted to it.

## 4. Measuring the price of debt and firm exposure

### *4.1. The price of corporate debt*

#### *4.1.1. Construction and validation*

For firm $i$ in year $t$ the implied borrowing cost is

$$r_{it} = \frac{|\text{Interest expense}_{it}|}{\frac{1}{2}\left(D_{it} + D_{i,t-1}\right)}, \tag{17}$$

with $D$ the sum of short- and long-term borrowings. Averaging the denominator prevents the ratio being driven mechanically by borrowing undertaken during the year. By Proposition 8, $\log r_{it}$ is contaminated only by an additive time effect under restatement, and we use logs throughout.

The measure is not calibrated to any policy series, so recovering one is a genuine test. Table 4 reports it.

Table 4: Median implied borrowing cost and the policy path

| Vietnam | | | Türkiye | | |
|---|---|---|---|---|---|
| Year | Implied | Lending rate | Year | Implied | CBRT policy |
| 2011 | 11.1% | 17.0% | 2022 | 17.8% | 9.0% |
| 2013 | 7.5% | 10.4% | 2023 | 25.4% | 42.5% |
| 2015 | 5.8% | 7.1% | 2024 | **31.2%** | **47.5%** |
| 2020 | 6.3% | 7.6% | 2025 | 25.8% | 38.0% |
| 2023 | 7.2% | 9.3% | | | |
| 2025 | 5.8% | — | | | |

Figure 1 plots the comparison. The Turkish series peaks in 2024, the year policy peaks. The Vietnamese series peaks in 2011 at the height of that

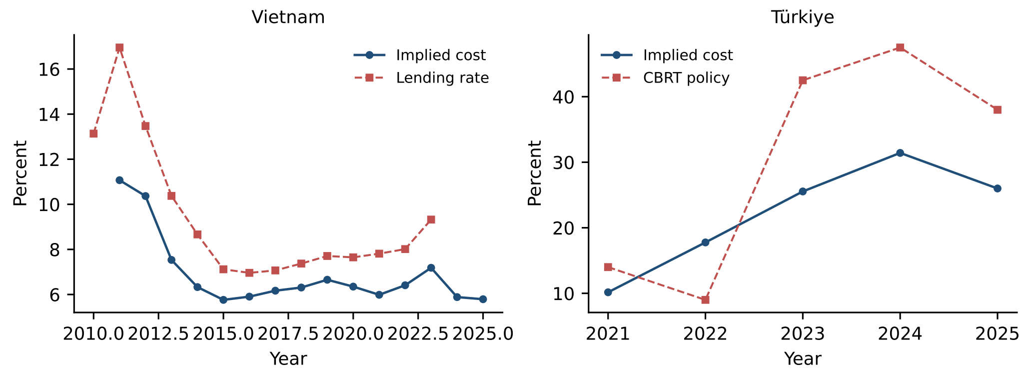


Figure 1: Median implied borrowing cost against the policy path. The measure is constructed from firm financial statements alone and is not calibrated to any policy series.

country's inflation crisis, falls monotonically to 5.8% by 2015, rises to 7.2% in the 2023 tightening and subsides. Median interest coverage moves in mirror image, reaching its Vietnamese sample minimum of 2.05 in exactly 2023.

#### *4.1.2. Pass-through*

Policy moved 38.5 points between end-2022 and the 2024 Turkish peak; realised borrowing costs moved 13.4 points, from 17.8% to 31.2%. By Corollary 9 this identifies a mean effective repricing share of $\bar{m} \approx 0.35$. Set against a median short-term debt share of 0.63, it implies that even nominally short debt repriced only partially. Over the same window median interest coverage halved, from 3.36 to 1.66. Vietnam's mild cycle moves costs 0.8 points, with coverage dipping from 3.11 to 2.05 and recovering to 2.70.

#### *4.1.3. A data hazard: bundled financial expense*

The Turkish field labelled *Financial Expenses* is the only interest-like item available before 2021 and is the natural candidate for the long history.

It should not be used as one: against the independent aggregator it runs roughly two and a half times larger, with a median relative difference of $-59.6\%$ and a correlation of only 0.495 on absolute values, against Vietnam's 0.991 and 0.00%.

The field bundles foreign-exchange revaluation, and in a currency crisis those losses dwarf interest. The consequence is not attenuation but a change in what is measured: built on the bundled field the Turkish implied cost peaks in **2018 and 2021**, the lira depreciation episodes; built on a clean interest figure it peaks in **2024**, with policy. A researcher using it to study monetary transmission would recover currency shocks and report them as interest-rate effects, with strong significance.

### *4.2. Measuring exposure*

#### *4.2.1. The required object is not disclosed*

Contractual repricing basis does not appear in published statements under either regime. Firms disclose borrowings by maturity and sometimes currency, but not by repricing terms. We therefore assess the two constructible substitutes.

#### *4.2.2. Revealed repricing sensitivity*

We estimate (6) firm by firm over a window ending before treatment. The FX term is included jointly rather than as a nuisance control, because Section 4.1 established that much of the movement in Turkish realised cost is currency revaluation.

On conventional diagnostics the estimates behave: 73% of Vietnamese firms and 67% of Turkish firms return a positive $\widehat{\theta}$, the mean Vietnamese estimate implies that a one-point rise in the lending rate raises a firm's cost of funds by roughly 0.79 points, and median in-sample $R^2$ is 0.28 in Vietnam and 0.34 in Türkiye. Propositions 2–6 supply the tests these diagnostics cannot pass, reported in Table 5.

Table 5: Split-half reliability of candidate exposure measures

| Measure | Türkiye | Vietnam |
|---|---|---|
| Repricing sensitivity $\widehat{\theta}$ | $\mathbf{-0.075}$ | $\mathbf{-0.082}$ |
| FX sensitivity $\widehat{\gamma}$ | +0.103 | +0.007 |
| Short-term debt share | +0.438 | +0.628 |
| Export share | +0.838 | — |

By Proposition 2 these correlations *are* the reliability ratio $\lambda$. Both are zero or negative, so by Corollary 3 the treatment estimand is attenuated to zero and carries no information about $\beta$. The measured balance-sheet characteristics in the lower rows behave as stable traits, so the failure is specific to regression-estimated quantities.

Table 6 applies Proposition 5 and delivers the paper's sharpest methodological finding. Sampling noise alone implies reliabilities of 0.996 and 0.898: a conventional power calculation, or an inspection of standard errors, would have judged the design comfortably feasible. The split-half correlations nonetheless force the permanent component to zero, attributing 99.4% of Vietnamese and 78.0% of Turkish dispersion to a transitory component.

The distinction matters for what one should do about it. Imprecision is remediable with more data; impermanence is not. A firm's estimated repricing sensitivity is real within the window in which it is measured and

Table 6: Variance decomposition of estimated repricing sensitivities

| | Vietnam | Türkiye |
|---|---|---|
| $\mathrm{Var}(\widehat{\theta})$ | $4.51 \times 10^{-3}$ | $1.74 \times 10^{-4}$ |
| Sampling component $\sigma_u^2$ | $1.62 \times 10^{-5}$ | $1.77 \times 10^{-5}$ |
| Reliability implied by sampling noise alone | 0.996 | 0.898 |
| Measured split-half correlation | −0.082 | −0.075 |
| Permanent component $\sigma_\theta^2$ | 0.000 | 0.000 |
| Transitory component $\sigma_\nu^2$ | $4.50 \times 10^{-3}$ | $1.56 \times 10^{-4}$ |
| Transitory share of dispersion | **99.6%** | **89.8%** |

does not persist beyond it, so it cannot serve as a pre-determined regressor for a treatment occurring later. Quarterly estimation would raise $T$ and shrink $\sigma_u^2$, which Table 6 shows was never the binding constraint.

*4.2.3. First-stage prediction*

The decisive test asks whether $\widehat{\theta}$ predicts what it purports to, reported in Table 7.

Table 7: Does estimated exposure predict the actual change in borrowing cost?

| | Türkiye (2022–24) | Vietnam (2022–23) |
|---|---|---|
| Coefficient on $\widehat{\theta}$ (standardised) | −0.036 | +0.001 |
| $p$-value | 0.27 | 0.54 |
| $R^2$ | 0.035 | 0.002 |
| Change, lowest tercile | +0.19 | +0.006 |
| Change, highest tercile | +**0.04** | +0.010 |

In Türkiye the most-exposed tercile records the *smallest* increase in realised cost. Median in-sample fit on the split-half windows is 0.49 in Vietnam and 0.66 in Türkiye against the null benchmark $k/(T-1) = 0.40$ of Proposition 6—the same order of magnitude as chance, and neither approaches the fit a genuine two-parameter relationship would typically produce.

#### *4.2.4. Balance-sheet proxies*

The measured characteristics are reliable but do not identify repricing. Regressing the actual change in borrowing cost on standardised pre-period characteristics yields $R^2$ below 0.02 for the short-term debt share, leverage, the current ratio, size and interest coverage, in both countries.

The short-maturity share illustrates the confound. It is simultaneously a measure of repricing exposure and of credit quality, since firms able to roll short-term paper cheaply are disproportionately the strongest borrowers. In Türkiye the two align and coverage falls monotonically across terciles; in Vietnam the credit-quality effect dominates and the ordering inverts, the most short-funded tercile being the healthiest in most years (5.02 against 3.45 in 2021). A proxy whose sign flips across two structurally similar markets is not measuring a single construct.

## 5. Identification, results and robustness

### *5.1. Identification*

#### *5.1.1. The restatement confound, empirically*

Table 8 reports the Turkish median balance-sheet aggregates.

Median leverage more than halves between 2019 and 2024 while equity-to-assets rises by over twenty points, in a period of severe tightening and

Table 8: Turkish median balance-sheet aggregates

| Year | Equity / assets | Debt / assets |
|---|---|---|
| 2017 | 0.472 | 0.217 |
| 2019 | 0.419 | 0.226 |
| 2021 | 0.462 | 0.189 |
| **2022** | **0.558** | **0.138** |
| **2023** | **0.614** | **0.111** |
| 2024 | 0.646 | 0.100 |
| 2025 | 0.619 | 0.117 |

depreciation during which no such deleveraging occurred. The break begins in 2022, consistent with the April 2022 classification and the restatement of comparatives.

Proposition 7 predicts that a design using stock-based outcomes will be inconsistent whenever $\mathrm{Cov}(\theta_i, \delta_i) \neq 0$. Estimating it with FX exposure as treatment returns exactly that pathology: coefficients that are highly significant and wrongly signed, with firms holding more foreign-currency debt appearing to *improve* after 2023 (implied rate $-0.108$, $p < 0.0001$; Altman $Z''$ (Altman, 2005) $+0.813$, $p = 0.004$; leverage $-0.049$, $p < 0.0001$); an event-study path growing monotonically as restatement compounds ($+0.32$, $+0.83$, $+1.18$ across 2023–2025); and large significant pre-treatment coefficients. The result has every outward mark of a strong causal finding and is an accounting artefact.

*5.1.2. The surviving specification*

Propositions 7 and 8 imply three restrictions. Outcomes must be flows: we retain the log implied rate, interest coverage, EBIT margin and log debt growth, and discard leverage, equity ratios, return on assets and the emerging-market Altman $Z''$ score of Altman (2005), each of which combines a monetary numerator with a non-monetary stock. Exposure must predate the restatement, and we measure it at 2021. And the pre-period must permit a test of parallel trends; because EBIT margin and debt growth require no interest figure, the bundling problem of Section 4.1 is irrelevant to them and the national source's history back to 2015 can be used. The estimating equations are (1) and its dynamic analogue

$$Y_{it} \;=\; \sum_{k\neq 2022} \beta_k\big(E_i \cdot \mathbf{1}\{t=k\}\big) \;+\; \alpha_i \;+\; \lambda_t \;+\; \varepsilon_{it}, \tag{18}$$

with 2022 omitted and standard errors clustered by firm (Petersen, 2009). Because treatment timing is common within country we avoid the negative-weighting and heterogeneity problems identified by Goodman-Bacon (2021), de Chaisemartin and D'Haultfœuille (2020), Callaway and Sant'Anna (2021) and Sun and Abraham (2021), so the two-way estimator remains interpretable here; following Roth (2022) we report full event paths rather than conditioning on a pre-trend test statistic.

*5.2. Results*

Four coefficients reach conventional significance. The event studies show none is a treatment effect; Table 10 reports the sharpest case.

Figure 2 makes the point visually. Firms with more short-term debt grow debt more slowly in every year from 2015 onward, and the coefficient at

Table 9: Main estimates: exposure × post, flow outcomes

| Country | Exposure / Outcome | Firms | Coef. | SE | $p$ |
|---|---|---|---|---|---|
| TR | Short-term share / implied rate | 466 | 0.018 | 0.016 | 0.27 |
| TR | Short-term share / coverage | 468 | −0.421 | 0.718 | 0.56 |
| TR | Short-term share / EBIT margin | 499 | −0.013 | 0.012 | 0.27 |
| TR | Short-term share / debt growth | 502 | −0.002 | 0.035 | 0.96 |
| TR | FX exposure / implied rate | 491 | −0.025 | 0.021 | 0.23 |
| TR | FX exposure / coverage | 498 | 0.176 | 0.733 | 0.81 |
| TR | FX exposure / EBIT margin | 547 | 0.023 | 0.011 | 0.04 |
| TR | FX exposure / debt growth | 536 | −0.055 | 0.028 | 0.05 |
| VN | Short-term share / implied rate | 522 | −0.002 | 0.001 | 0.11 |
| VN | Short-term share / coverage | 521 | −0.355 | 0.497 | 0.48 |
| VN | Short-term share / EBIT margin | 522 | −0.016 | 0.004 | 0.00 |
| VN | Short-term share / debt growth | 522 | 0.059 | 0.019 | 0.00 |

Table 10: Event study: Turkish short-maturity share on debt growth (base 2022)

| Year | Coef. | SE | Year | Coef. | SE |
|---|---|---|---|---|---|
| 2015 | $-0.138^{**}$ | 0.068 | 2021 | $-0.232^{***}$ | 0.066 |
| 2016 | $-0.187^{***}$ | 0.065 | 2022 | base | — |
| 2017 | $-0.162^{***}$ | 0.062 | **2023** | $\mathbf{-0.220}^{***}$ | 0.069 |
| 2018 | $-0.189^{***}$ | 0.072 | 2024 | −0.069 | 0.076 |
| 2019 | $-0.252^{***}$ | 0.070 | 2025 | $-0.191^{***}$ | 0.064 |
| 2020 | $-0.211^{***}$ | 0.057 | | | |

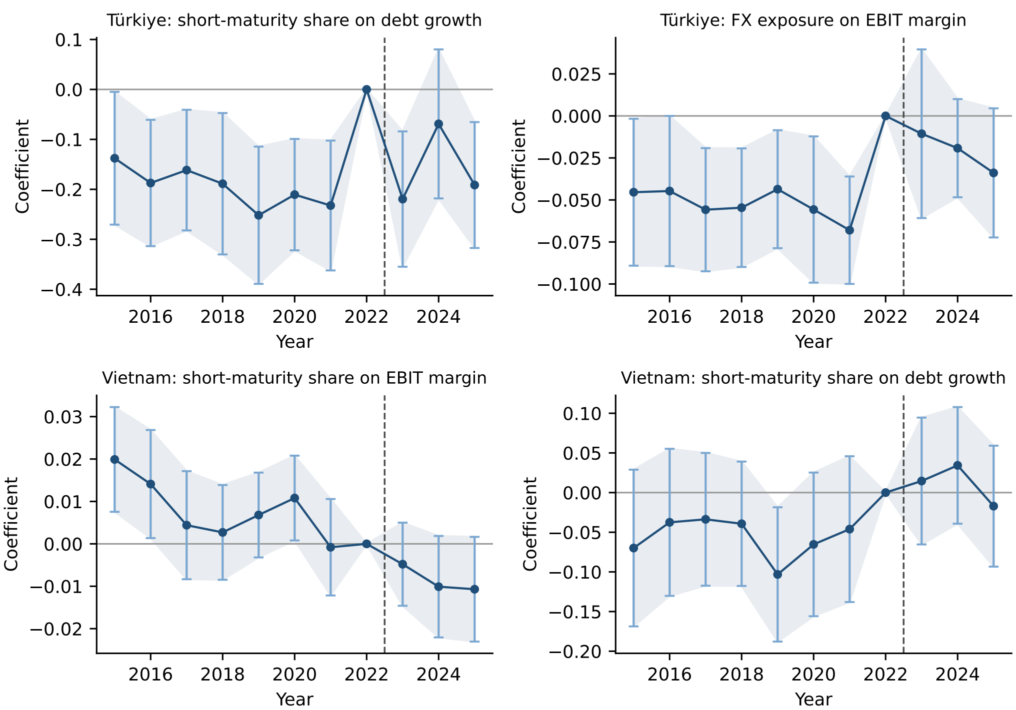


Figure 2: Event studies of exposure interacted with year, with firm and year effects and 95% confidence intervals clustered by firm. The dashed line marks the treatment date. Coefficients are non-zero well before treatment and do not shift at it.

treatment is statistically indistinguishable from those seven years earlier. The Vietnamese EBIT-margin result has the same character, drifting from +0.020 ($p < 0.01$) in 2015 to $-0.011$ ($p < 0.10$) in 2025 and passing smoothly through the treatment year. These are persistent differences between firm types, not responses to monetary policy. What a conventional two-year window would report is the difference between the 2022 base and the 2023–25 average—a difference that exists, is significant, and means nothing causal.

### *5.3. Cross-country evidence: dose-response and pooled exposure*

The two-country design identifies the exposure gradient cleanly but leaves open whether its absence is a Türkiye-and-Vietnam idiosyncrasy or a general property of listed emerging-market firms. We extend the price validation of Section 4 and the exposure test of this section to the five-country, 2021–2025 sample described in Section 3.2, spanning tightening doses from Türkiye's 33.5 percentage points to Vietnam's 0.8.

Table 11: Dose-response across five countries, 2022 to 2023–24

| Country | Dose (pp) | Firms | $\Delta$ Implied rate (pp) | ICR pre | ICR post |
|---|---|---|---|---|---|
| Türkiye | 33.50 | 559 | 10.59 | 3.36 | 2.45 |
| Poland | 5.00 | 521 | 1.59 | 4.72 | 2.91 |
| Israel | 4.65 | 274 | 1.90 | 4.81 | 3.29 |
| South Korea | 2.50 | 290 | 1.56 | 6.19 | 4.16 |
| Vietnam | 0.80 | 570 | 0.13 | 2.91 | 2.23 |

The change in implied borrowing cost correlates with policy dose at $r = 0.996$ across the five countries, and median interest coverage falls in every single one, from Türkiye's 33.5-point tightening down to Vietnam's 0.8. Figure 3 plots the relationship.

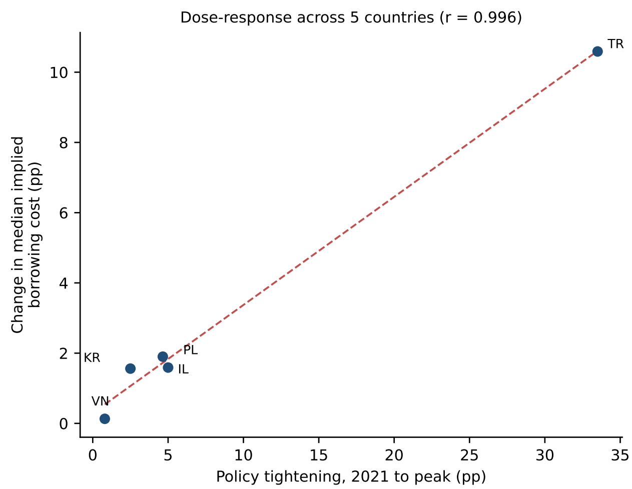


Figure 3: Change in median implied borrowing cost against policy tightening dose, five countries, 2022 to 2023–24. The measure is constructed from firm statements alone.

A correlation of 0.996 on five points warrants an immediate caveat rather than a boast, and we give it here rather than let the reader discover it independently. With $n = 5$, a very high correlation is close to mechanical: four of the five points already sit on an economically reasonable line, and a fifth far enough from the others (Türkiye) will produce a high $r$ regardless of the details of the middle three. We therefore read Table 11 qualitatively—interest coverage deteriorates in every country, in rough proportion to tightening severity, and the pattern is consistent with the four additional markets assembled but not carried into this table (see the footnote to Section 1)—rather than treating 0.996 as a precisely estimated population parameter. The qualitative claim is the one this design can support; the correlation coefficient illustrates it and should not be read as more than that.

We then pool all five countries and estimate the triple interaction

$$Y_{ict} \;=\; \beta_1\big(E_i P_t\big) \;+\; \beta_2\big(E_i P_t \cdot \mathrm{Dose}_c\big) \;+\; \alpha_i \;+\; \lambda_{ct} \;+\; \varepsilon_{ict}, \tag{19}$$

with $E_i$ the pre-period short-term debt share (the only exposure proxy constructible on a four-year panel), $\mathrm{Dose}_c$ the country's standardised tightening severity, and $\lambda_{ct}$ country-by-year effects. If the exposure gradient were real and simply underpowered in Section 5.2's two-country test, $\beta_2$ should be positive and significant on distress outcomes as dose rises. Table 12 reports the estimates.

Table 12: Pooled exposure × dose interaction, five countries

| Outcome | Firms | $\beta_1$ ($p$) | $\beta_2$ ($p$) |
|---|---|---|---|
| Interest coverage | 598 | −0.979 (0.071) | 0.022 (0.932) |
| EBIT margin | 598 | −0.008 (0.136) | −0.009 (0.418) |
| Implied rate | 604 | 0.008 (0.073) | 0.032 (0.048) |

The dose interaction is null on both distress outcomes: interest coverage ($p = 0.932$) and EBIT margin ($p = 0.418$) show no evidence that the exposure gradient steepens with tightening severity. It is significant only for the implied rate itself ($p = 0.048$)—the mechanical restatement, at firm level, of the aggregate result already established in Table 11, not a new finding about cross-sectional exposure. Scaling the test across a range of monetary shocks that spans more than an order of magnitude produces the same conclusion as the two-country design: the price of debt responds to dose, the firm-level exposure gradient does not.

We emphasise the limits of this extension. The short-term debt share is the proxy shown in Section 4.2 to conflate repricing exposure with credit quality, and the four-year panel available for Poland, Israel and South Korea is, by the feasibility condition of Proposition 4, too short to support the reliability-tested repricing beta of Section 4.2. This subsection therefore corroborates the two-country null under a cruder exposure measure and a wider dose range; it does not repeat the decade-long identification test, which remains specific to Türkiye and Vietnam. Nor have these three countries been cross-validated against an independent second source in the manner of Section 4's reconciliation exercise, which was performed only for Vietnam and Türkiye; their statement quality is assumed rather than independently verified, a limitation the two-country panels do not share.

### *5.4. Robustness*

*Placebo treatment..* Assigning a false treatment at 2018 and restricting to 2015–2021 returns nothing for Vietnam: $-0.791$ ($p = 0.19$) on interest coverage and $+0.004$ ($p = 0.51$) on EBIT margin. The design is not mechanically generating significance, which supports reading Section 5.2's significant coefficients as genuine pre-existing trends.

*Sample-size stability..* The Turkish estimates were computed at three sample sizes as collection proceeded. At 83 firms the specification returned $+0.886$ on the Altman $Z''$ score ($p = 0.045$) with significant pre-trends; at 158 firms both vanished; at 585 firms the estimates are those of Table 9. Roughly a third of the eventual sample produced a significant wrong-signed result with apparent pre-trend violation.

*Alternative specifications..* Results are unchanged under winsorisation at 1%/99% rather than 2%/98%; exposure measured at 2020 or as the 2020–21 mean; exclusion of the smallest asset tercile; and exclusion of 2022 to guard against restated comparatives.

*What would overturn the null..* Table 6 is explicit that quarterly estimation would shrink $\sigma_u^2$, which was not the binding constraint, and would leave $\sigma_\nu^2$ untouched. Portfolio sorting would average away the transitory component at the cost of the heterogeneity that motivates the exercise. Machine-readable disclosure of repricing basis would remove the problem entirely.

## 6. Discussion and conclusion

### 6.1. Implications for practice

*Validate the price measure against the policy path.* A measure that cannot reproduce a known aggregate should not be used to estimate an unknown cross-section; this test is what exposed the bundled-expense problem.

*Report split-half reliability whenever exposure is estimated.* By Proposition 2 it is the attenuation factor itself, not merely a diagnostic correlated with it, and by Proposition 5 it separates imprecision from impermanence. Standard errors and power calculations cannot: in our data they indicated reliabilities of 0.78–0.99 where the true permanent share was zero.

*In inflation-accounting regimes, prefer flows.* By Proposition 7 the contamination enters as an interaction and fixed effects do not address it; by Proposition 8 ratios of flows escape it.

*Extend the pre-period until parallel trends can genuinely be tested.* Restricting to outcomes with long history revealed that our significant coefficients were decade-long trends; a two-year window would have produced a publishable false positive.

### 6.2. Conclusion

Published financial statements support a genuine, validated measure of the price of corporate debt, and it recovers monetary policy paths in a severe Turkish tightening and a mild Vietnamese one. Using it we document pass-through of roughly 35%, a halving of median Turkish interest coverage, and a data field that conflates currency shocks with interest costs.

Statements do not support a measure of which firms are contractually exposed to repricing. Our framework explains why in a form that is testable rather than rhetorical: the estimand is attenuated by the reliability of the exposure proxy, that reliability is identified by a split-half correlation which is zero in both countries, and the dispersion in estimated sensitivities is transitory rather than permanent—so the failure is impermanence, which more data cannot cure.

The implication is not that the channel fails to operate; our aggregate evidence is consistent with its operating forcefully in Türkiye. It is that the cross-sectional question—*which* firms bear the adjustment—requires contractual data. Researchers without registry access should direct effort toward the aggregate and price-side questions statements can answer, and toward making repricing terms disclosable, rather than toward proxies that cannot carry the weight placed on them.

## Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used generative AI tools to support the writing of the manuscript. The author reviewed, revised and rewrote all AI-assisted content as necessary and takes full responsibility for the content of the published article.

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